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CHENNAI

## Final Assessment Test(FAT) - NOV/DEC 2025

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| Programme | B.Tech. | Semester | Fall Semester 2025-26 |
| Course Code | BCSE355L | Faculty Name | Prof. Malini A |
| Course Title | Cloud Architecture Design | Slot | B2+TB2 |
| | | Class Nbr | CH2025260101477 |
| Time | 3 hours | Max. Marks | 100 |

**Instructions To Candidates**

- Write only your registration number in the designated box on the question paper. Writing anything elsewhere on the question paper will be considered a violation.

**Course Outcomes**

CO1: Demonstrate an in-depth understanding of AWS Cloud architecture, services,and design patterns.

CO2: Apply security and compliance measures effectively in AWS architectures using encryption, access controls, and monitoring.

CO3: Optimize cost and performance by selecting appropriate AWS services and utilizing cost-effective resource management strategies.

CO4: Configure and manage advanced AWS networking features, storage solutions, database technologies, and compute resources.

### Section - I

### Answer all Questions (7 × 10 Marks)

01. A multinational media corporation is preparing to launch a video streaming platform that is latency-sensitive across continents. The solution must ensure high availability, comply with local data residency regulations, and deliver seamless performance.

(a) Examine how AWS Regions, Availability Zones, and Edge Locations work together to support the architectural requirements of a global streaming service. Explain the roles that each component plays in achieving fault tolerance, performance optimization, and regulatory compliance. **(6 Marks)**

(b) To make sure internal teams understand their security duties, the company's is examining the AWS Shared Responsibility Model. List the four primary security responsibilities that are under the customer. Discuss about how these duties affect data, application, and infrastructure security in the context of a media streaming **(4 Marks)**

**[10] (CO1/K2)**

02. Cambridge International University manages a large research computing environment across multiple departments. Researchers collaborate on projects that generate large datasets such as simulation results, lab reports, and academic publications. These datasets require concurrent access from multiple compute instances and labs, with high throughput for data-intensive computations. The university also requires scalable, secure, and cost-effective shared file storage that integrates with cloud compute resources while enforcing strict access control for sensitive research data.

(a) Design a managed cloud file system architecture for Cambridge International University that supports concurrent access, high throughput, and scalable storage for research datasets. (5 Marks)

(b) Explain how Cambridge International University can implement security and access control for the managed cloud file system to protect sensitive research data. (5 Marks )

**[10] (CO2/K4)**

03. A hospital network is developing a real-time patient monitoring system where vital signs (e.g., heart rate, blood pressure, oxygen saturation) from wearable devices are ingested into Amazon DynamoDB. These updates must initiate immediate alerts for anomalies and be replicated to a centralized Amazon Aurora database for analytics, reporting, and long-term regulatory storage. During high-demand periods, the system processes thousands of patient updates per second. Key requirements includes:

a) Delivering alerts to medical staff within 2 seconds of detecting abnormal vitals. Design an architecture using DynamoDB Streams and Aurora to handle real-time alerting and durable storage for patient vitals. [3 Marks]

b) Ensuring scalable, durable storage for historical data to support compliance and analysis. Identify potential performance bottlenecks in high-throughput vitals ingestion and suggest optimizations to maintain scalability and low latency. [4 Marks]

c) Achieving cost-effectiveness with high availability and minimal latency. Explain how to implement failure handling, data integrity mechanisms, and compliance with healthcare regulations such as Health Insurance Portability and Accountability Act (HIPAA) and General Data Protection Regulation (GDPR). [3 Marks]

**[10] (CO2/K4)**

04. A growing e-commerce company has recently migrated all its workloads to AWS, but after six months, the finance team reports that monthly AWS costs are rising unpredictably, occasionally exceeding budget forecasts by 35%. The engineering team mainly uses on-demand EC2 instances for development and testing, and several teams generate large amounts of application logs which are stored in S3 Standard without lifecycle policies. There is no tagging strategy or system to allocate costs by project or team, causing a lack of financial accountability. Developers admit they do not fully understand how AWS billing works.

(i) Summarize on why unpredictable cloud costs are often a challenge compared to traditional on-premise environments and describe two key cost-driving practices evident in this scenario. (5 Marks)

(ii) Examine on the AWS Cost Management tools that the company should use to gain insight into hidden costs and spending patterns, and propose a governance strategy that balances developer innovation with fiscal responsibility. (5 Marks)

**[10] (CO3/K4)**

05. Abdul works at StreamWave Technologies as a cloud architect responsible for enhancing the company's global media processing and streaming system. The platform runs intensive workloads such as video transcoding, thumbnail generation, and content metadata extraction on Amazon EC2 instances.StreamWave aims to improve scalability, performance, and fault tolerance while optimizing storage costs and reliability across its multi-region architecture.

a) Design a scalable and high-performance storage architecture using block storage and file storage for StreamWave Technologies' media workloads (5 marks)

b) Analyze how the combination of block storage and file storage enhances performance, availability, and cost efficiency for StreamWave Technologies (5 marks)

**[10] (CO3/K3)**

06. CloudSecure Pvt. Ltd. manages server infrastructure for multiple clients. During security incidents such as DDoS attacks or unauthorized logins, their monitoring tool logs events but alerts often reach admins hours later, delaying response. To improve responsiveness, the company integrates Amazon SNS (Simple Notification Service) into its cloud monitoring workflow.

The plan includes:

- Creating SNS Topics for publishing security alerts.
- Adding multiple subscribers Email, SMS, AWS Lambda, and HTTPS endpoints.
- Using Amazon CloudWatch alarms as publishers that trigger notifications.
- Applying Message Attribute Filters so that each team receives only relevant alerts.
- Optionally connecting SQS Queues for reliable message persistence.

a) Apply your understanding of event-driven architecture to describe the end-to-end alerting workflow of CloudSecure's system using Amazon SNS. Identify the roles of publishers, topics, and subscribers. (5 Marks)

b) Analyze how this new SNS-based design enhances responsiveness, scalability, and integration compared to the old manual alerting process. (3 Marks)

c) Evaluate two implementation challenges that may arise while adopting SNS in this context, and propose suitable solutions. (2 Marks)

**[10] (CO4/K3)**

07. a) A renewable energy research Centre operates in a remote location where internet connectivity is very limited. The company collects 10 TB of daily sensor data from solar panels and weather monitoring devices. The Centre wants to process part of the data locally for quick insights and then migrate all data to AWS S3 for centralized analysis once every three weeks.

Explain how AWS Snowball Edge enables local data processing and later transfer to AWS ( 5 marks)

b) A global media company maintains an on-premises data center with 10 petabytes of video archives. They plan to close the data center and migrate all archives to Amazon S3 Glacier Deep Archive for long-term, low-cost storage. But transferring this massive data volume through the internet would take several months and be too costly.

Explain how AWS Snowmobile can be used to perform this migration efficiently.( 5 marks)

**[10] (CO4/K2)**

## Section - II

**Answer all Questions (2 × 15 Marks)**

08. The team cannot guarantee reliable communication between the production Virtual Private Cloud (VPC) in Account A and the development VPC in Account B, leading to delays in data synchronization. A security audit has indicated that EC2 instances in the production VPC are permitting unauthorized inbound traffic, posing a threat to security.

(a) Identify one VPC configuration scenario that could cause failure in cross-account and cross-region communication between Account A and Account B. Describe how the misconfiguration affects connectivity. (4 Marks)

(b) Analyze a second VPC configuration scenario that could cause communication failure between Account A and Account B. Illustrate its impact on data synchronization. (4 Marks)

(c) Suggest and describe a solution to enable secure communication between the two VPCs. (7 Marks)

**[15] (CO1/K4)**

09. The military system is implementing measures to ensure the security of all computers that safeguard against potential threats from foreign nations. This involves the utilization of Amazon Web Services (AWS) for data storage, specifically within S3 instances, which provide a robust and scalable solution for managing sensitive information. To enhance security protocols, the system establishes IAM (Identity and Access Management) user permissions tailored to the specific roles and responsibilities of personnel. They are developed Key management Service (KMS). This approach not only streamlines access control but also fortifies the overall defense strategy by ensuring that individuals have the appropriate level of access based on their designated functions within the organization.

a) Justify the best solution for optimizing KMS performance when multiple encryption/decryption requests occur per second.(5 marks)

b) Examine how the military team can recover access after IAM permissions for a KMS key are accidentally deleted (5 marks)

c) Determine which AWS service can identify who used a KMS key and when it happened.(5 marks)

**[15] (CO2/K4**

**BL-Bloom's Taxonomy Levels - (K1-Remembering,K2-Understanding,K3-Applying,K4-Analysing,K5-Evaluating,K6-Creating)**

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